

Lecture 13/Module 2

Transient Analysis of First and Second Order Circuit

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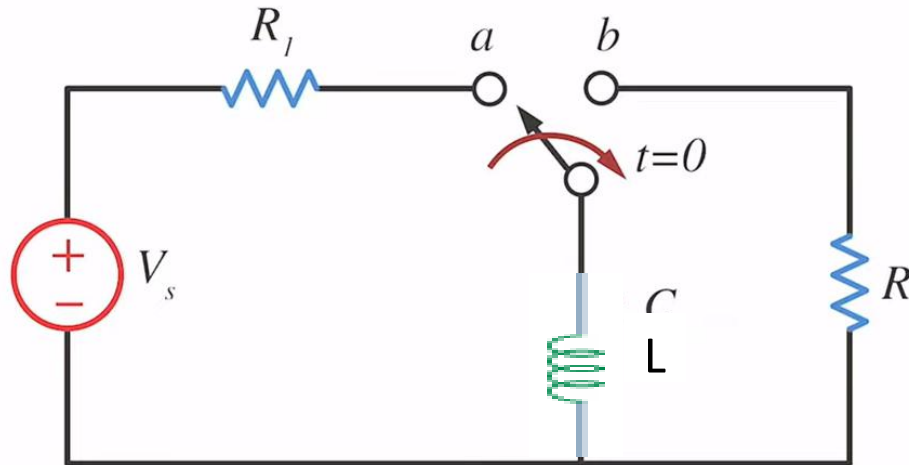


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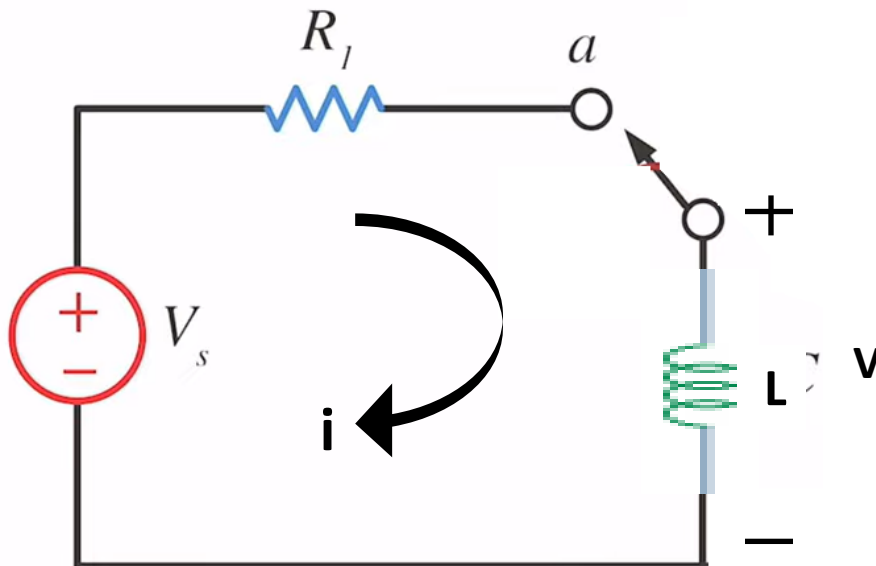
Today's Topics

- Source Free RL Circuit – Natural Response
- Exercise
- Singularity Function

Initial Condition of RL Circuit



Before $t = 0$ OR $t < 0$

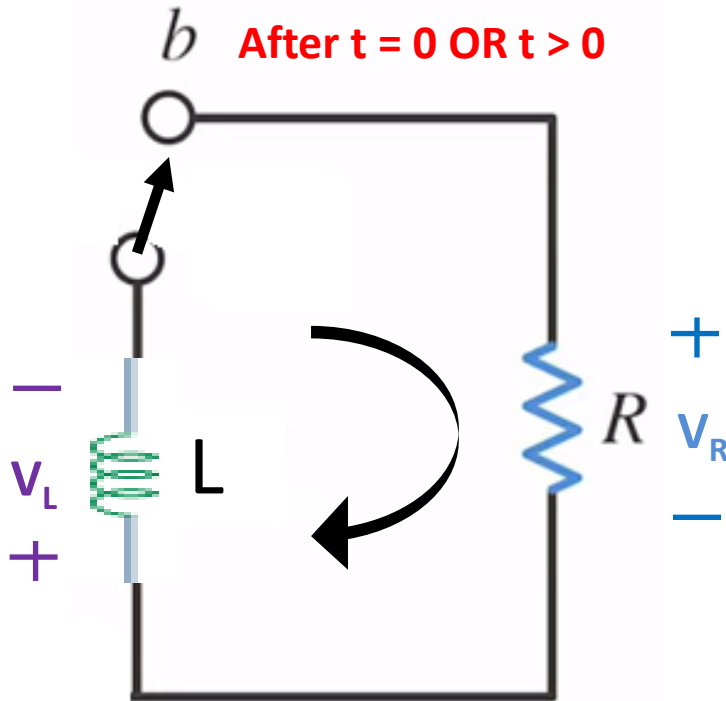


Initial Condition

$$V = L \frac{di}{dt}$$

$$i = \frac{1}{L} \int V dt = I_0$$

Natural (Source Free) Response of RL Circuit



KVL in mesh

$$V_R + V_L = 0$$

By definition, $v_L = L \, di/dt$ and $v_R = iR$. Thus,

$$L \frac{di}{dt} + Ri = 0$$

$$\frac{di}{dt} + \frac{R}{L}i = 0$$

This is a first-order differential equation,

Rearranging terms and int

$$\int_{I_0}^{i(t)} \frac{di}{i} = - \int_0^t \frac{R}{L} dt$$

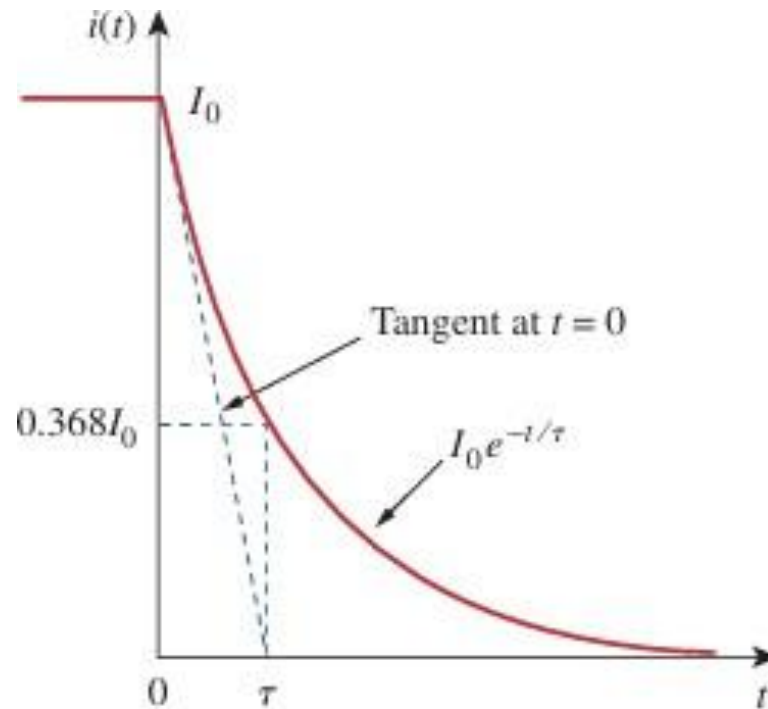
$$\ln i \Big|_{I_0}^{i(t)} = - \frac{Rt}{L} \Big|_0^t \Rightarrow \ln i(t) - \ln I_0 = - \frac{Rt}{L} + 0$$

The Source-Free RL Circuit

Taking powers of e produces,

$$i(t) = I_0 e^{-Rt/L}$$

The equation shows that the natural response of the RL circuit is an exponential decay of the initial current.



The natural response of a circuit refers to the behavior (in terms of voltages and currents) of the circuit itself, with no external sources of excitation.

The Source-Free RL Circuit

The time constant for the RL circuit is

$$\tau = \frac{L}{R}$$

Thus,

$$i(t) = I_0 e^{-t/\tau}$$

τ has the units of Seconds. The smaller the time constant of a circuit, the faster the rate of decay of the response. The larger the time constant, the slower the rate of decay of the response.

Voltage across the resistor is $v_R(t) = iR = I_0 R e^{-t/\tau}$

The power dissipated in the resistor is $p = v_R i = I_0^2 R e^{-2t/\tau}$

The energy absorbed by the resistor is

$$w_R(t) = \int_0^t p(\lambda) d\lambda = \int_0^t I_0^2 R e^{-2\lambda/\tau} d\lambda = -\frac{\tau}{2} I_0^2 R e^{-2\lambda/\tau} \Big|_0^t, \quad \tau = \frac{L}{R}$$

$$w_R(t) = \frac{1}{2} L I_0^2 (1 - e^{-2t/\tau})$$

The Source-Free RL Circuit

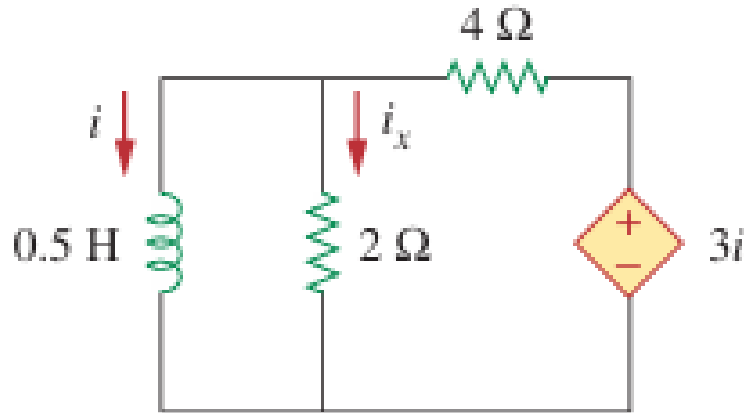
Notice that as t tends to infinity, w tends to $\frac{1}{2} LI_0^2$, which is the same as the energy initially stored in the inductor.

The energy that was initially stored in the inductor is eventually dissipated in the resistor.

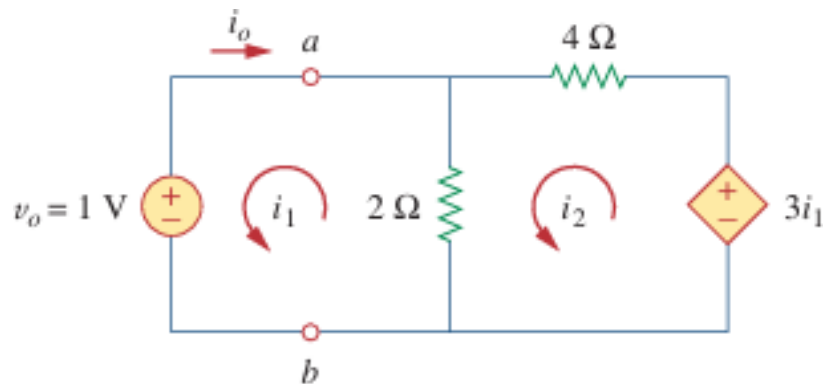
When a circuit has a single inductor and several resistors and dependent sources, the Thevenin equivalent can be found at the terminals of the inductor to form a simple RL circuit. Also, one can use Thevenin's theorem when several inductors can be combined to form a single equivalent inductor.

Tutorial 1

1. Assuming $i(0) = 10 \text{ A}$, calculate $i(t)$, and $i_x(t)$ in the circuit.



Because of the dependent source, we insert a voltage source with $v_o = 1 \text{ V}$ at the inductor terminals a-b.



Applying KVL to the two loops,

$$2(i_1 - i_2) + 1 = 0 \quad \Rightarrow \quad i_1 - i_2 = -\frac{1}{2}$$

$$6i_2 - 2i_1 - 3i_1 = 0 \quad \Rightarrow \quad i_2 = \frac{5}{6}i_1$$

Hence, $R_{\text{eq}} = R_{\text{Th}} = \frac{v_o}{i_o} = \frac{1}{3} \Omega$

$$i_1 = -3 \text{ A}, \quad i_o = -i_1 = 3 \text{ A}$$

The time constant is

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{\frac{1}{2}}{\frac{1}{3}} = \frac{3}{2} \text{ s}$$

Thus, the current through the inductor is

$$i(t) = i(0)e^{-t/\tau} = 10e^{-(2/3)t} \text{ A}, \quad t > 0$$

The voltage across the inductor is, $v = L \frac{di}{dt} = 0.5(10) \left(-\frac{2}{3} \right) e^{-(2/3)t} = -\frac{10}{3} e^{-(2/3)t} \text{ V}$

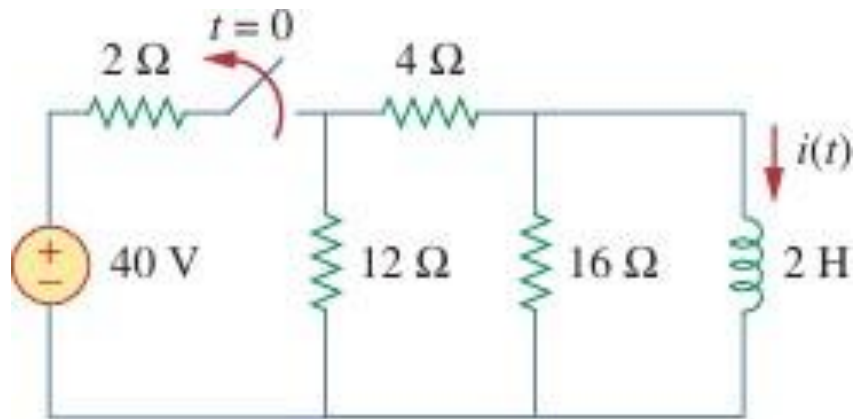
Since the inductor and the resistor are in parallel,

$$i_x(t) = \frac{v}{2} = -1.6667 e^{-(2/3)t} \text{ A}, \quad t > 0$$

Tutorial 2

2. The switch in the circuit has been closed for a long time. At the $t = 0$, the switch is opened.

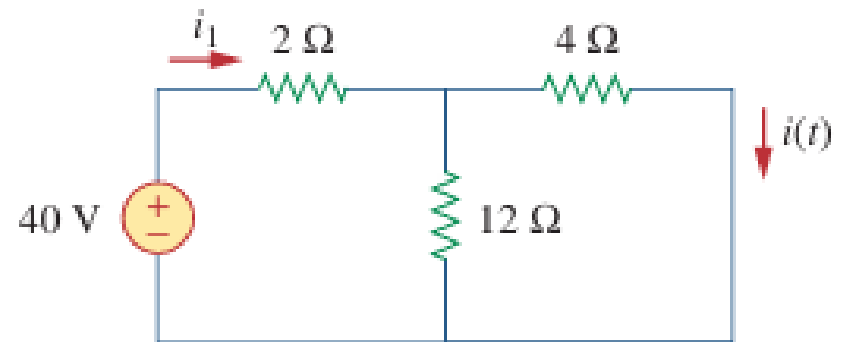
Calculate $i(t)$ for t greater than zero.



$$\frac{4 \times 12}{4 + 12} = 3 \Omega$$

$$i_1 = \frac{40}{2 + 3} = 8 \text{ A}$$

When the switch is closed, and the inductor acts as a short circuit to dc. The 16Ω resistor is short-circuited;



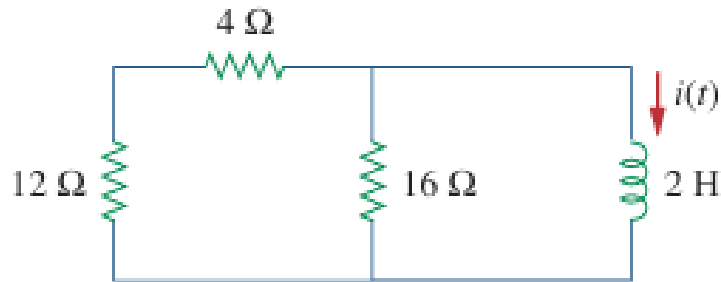
We obtain $i(t)$ using current division

$$i(t) = \frac{12}{12 + 4} i_1 = 6 \text{ A}, \quad t < 0$$

Since the current through an inductor cannot change instantaneously.

$$i(0) = i(0^-) = 6 \text{ A}$$

When the switch is open at t greater than 0 and the voltage source is disconnected. We now have the source-free RL circuit,



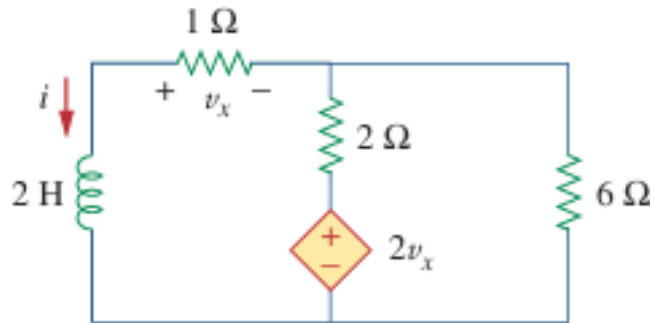
$$R_{\text{eq}} = (12 + 4) \parallel 16 = 8 \Omega$$

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{2}{8} = \frac{1}{4} \text{ s}$$

$$i(t) = i(0)e^{-t/\tau} = 6e^{-4t} \text{ A}$$

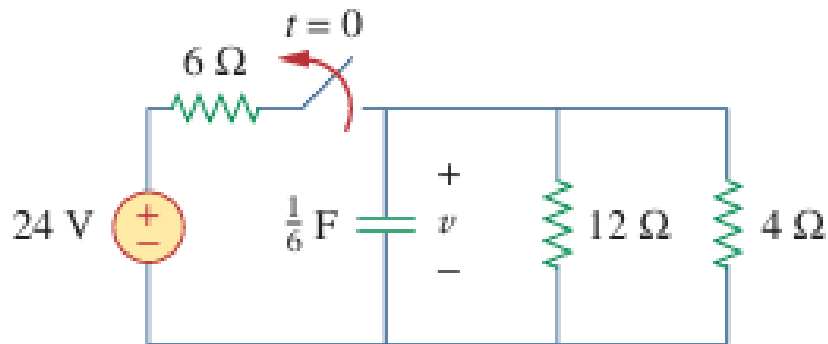
Tutorial 3

3. Find I and v_x in the circuit. Let $i(0) = 12A$



Tutorial 4

4. If the switch opens at $t = 0$, find $v(t)$ for $t \geq 0$ and $w_c(t)$.



Singularity Functions

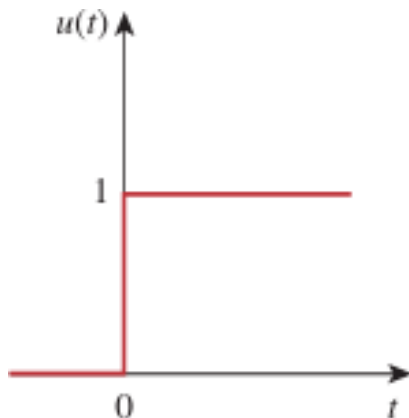
Singularity functions are also called switching functions.

Singularity functions are functions that either are discontinuous or have discontinuous derivatives.

The three most widely used singularity functions in circuit analysis are the **unit step**, the **unit impulse**, and the **unit ramp** functions.

Unit Step:

The unit step function $u(t)$ is 0 for negative values of t and 1 for positive values of t .



Mathematical equation,

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

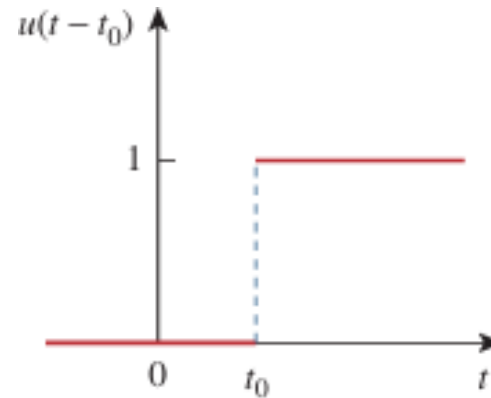
Singularity Functions

The unit step function is undefined at $t = 0$ where it changes abruptly from 0 to 1.

If the abrupt change occurs at $t = t_0$ (where t_0 is *greater than zero*) instead of $t = 0$ the unit step function becomes,

$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

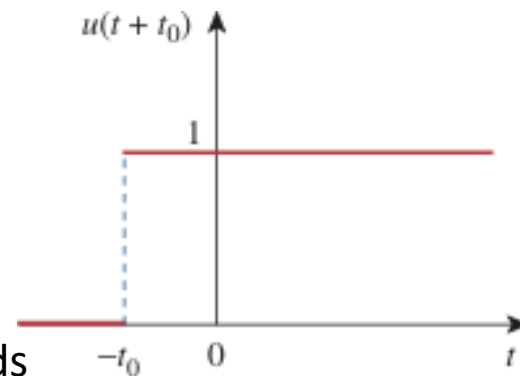
This means the same as saying that the unit step is delayed by t_0 seconds.



If the change is at $t = -t_0$, the unit step function becomes,

$$u(t + t_0) = \begin{cases} 0, & t < -t_0 \\ 1, & t > -t_0 \end{cases}$$

This means that the step has advanced by t_0 seconds



Singularity Functions

We use the step function to represent an abrupt change in voltage or current, like the changes that occur in the circuits of control systems and digital computers.

For example, if the voltage $v(t)$ is mathematically expressed as,

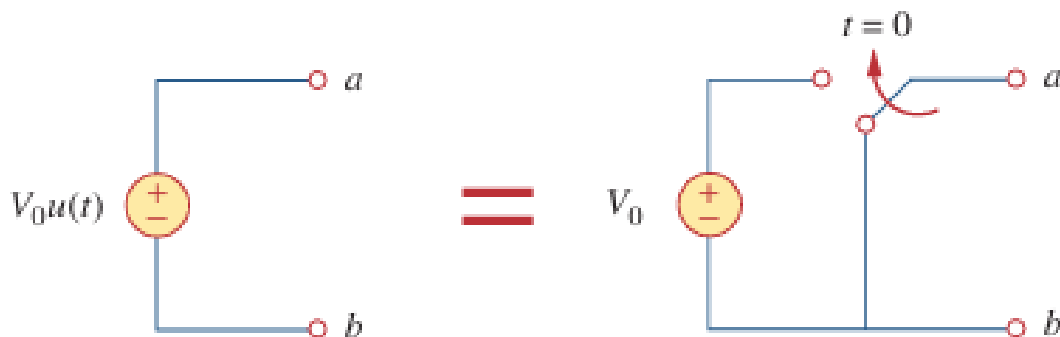
$$v(t) = \begin{cases} 0, & t < t_0 \\ V_0, & t > t_0 \end{cases}$$

This may be expressed in terms of the unit step function as,

$$v(t) = V_0 u(t - t_0)$$

If we let $t_0 = 0$, then $v(t)$ is simply the step voltage $V_0 u(t)$.

A voltage source $V_0 u(t)$ and its equivalent is shown as,

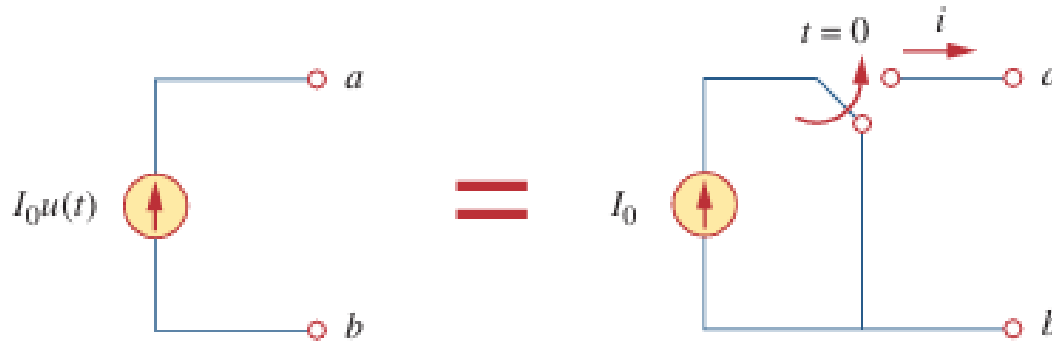


terminals a-b are short circuited for t less than zero and $v = 0$.

For t greater than 0, $v = V_0$.

Singularity Functions

Similarly, a current source of $I_0 u(t)$ is shown,



Notice that for t less than zero, there is an open circuit ($i = 0$), and that $I = I_0$ for t greater than zero.

Delta Function/Impulse function:

The derivative of the unit step function $u(t)$ is the **unit impulse function** denoted by $\delta(t)$. The unit impulse function—also known as the **delta function**.

The unit impulse function $\delta(t)$ is zero everywhere except at $t = 0$, where it is undefined.

$$\delta(t) = \frac{d}{dt}u(t) = \begin{cases} 0, & t < 0 \\ \text{Undefined}, & t = 0 \\ 0, & t > 0 \end{cases}$$

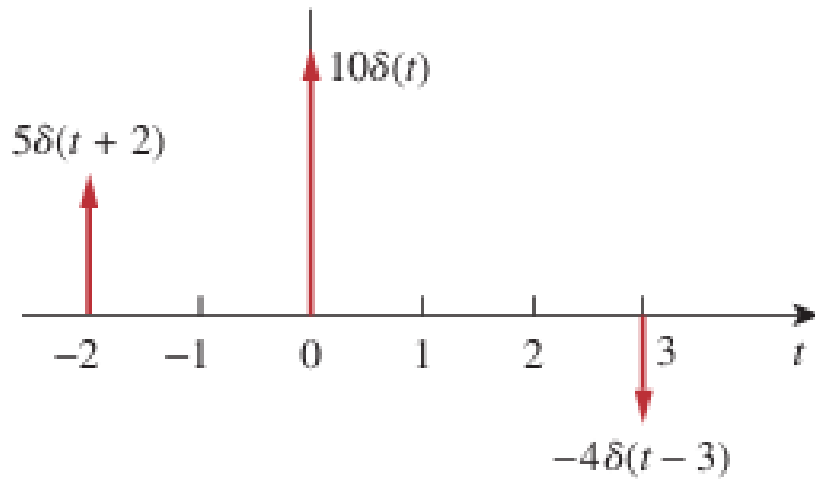
Singularity Functions

Impulsive currents and voltages occur in electric circuits as a result of switching operations or impulsive sources. Although the unit impulse function is not physically realizable (just like ideal sources, ideal resistors, etc.), it is a very useful mathematical tool.

It may be visualized as a very short duration pulse of unit area,
The unit area is known as the strength of the impulse function.

$$\int_{0^-}^{0^+} \delta(t) dt = 1$$

When an impulse function has a strength other than unity, the area of the impulse is equal to its strength. For example an impulse function has an area of 10.



Thank You